A Quick Look at Open Source Databases for Mobile App Development

Data is the life blood of the IT world. It is generated in great volumes in the course of our everyday lives. Efficient data management calls for the use of database management systems (DBMSs). This article presents readers with a selection of popular open source DBMSs.

Organisations today have a wide choice when it comes to choosing efficient database management systems (DBMSs). Various IT giants like Oracle, Microsoft, IBM and SAP offer high-end solutions that address all sorts of database requirements. Even newer IT companies like Amazon, Google and Rackspace have entered the fray, offering cloud-related features with regard to databases. According to Gartner analyst, Merv Adrian, 92.1 per cent of DBMS revenue comes from commercial software. Yet, new data is finding its way to open source databases as developers are opting for more convenience.

Open source cloud computing and database adoption is today widespread, as these databases provide huge benefits and are becoming a strong solution for every data management need within organisations. The DBMS space has been considerably impacted by the open source software (OSS) movement. The adoption of MySQL, MariaDB and other databases is saving organisations a lot of money, which would have otherwise been spent on licences and maintenance. Over recent years, various open source database projects have attained the maturity stage. In addition to cost savings, open source databases offer very easy tweaking, scaling and, overall, are feature-rich.

Apart from providing a solid backbone to enterprises in storing, retrieving and manipulating regular data during typical day-to-day operations, open source databases are also becoming a strong choice for mobile app developers.

Mobile operating systems have gained much popularity. Android and iOS have surpassed Windows OS as the most used operating systems in the world. As mobile OSs and mobile hardware are becoming more powerful, with mobiles becoming fully functional smart computing devices, data computation from mobiles is also increasing.

Databases are regarded as the most obvious way of storing and manipulating data. In the beginning, databases on mobile apps were handled on the server side or on the cloud. Mobile devices only communicated with them via the network. However, with the passage of time, to make the applications more responsive and less network dependent, offline databases have been gaining popularity. Nowadays, mobile apps store entire databases on the phone or have the option of keeping the data on the cloud, with synchronisation once a day, depending on the network connectivity. This leads to faster and more responsive applications.

Mobile databases

A mobile database is one that a mobile computing device can connect to over a wireless network. Mobile databases can further be classified as those that are physically separate from the central database server, those that reside on mobile